

DSP Short Term Time Domain Processing of Speech

S. R. Mahadeva Prasanna

Professor, Dept. of DSAI, IIIT Dharwad

prasanna@iiitdwd.ac.in

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Some Good Books in Speech Processing

- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- L.R. Rabiner and R.W. Schafer, "Digital Processing of Speech Signals", Pearson Education, Delhi, India, 2004
- J. R. Deller, Jr., J. H. L. Hansen and J. G. Proakis, "Discrete-Time Processing of Speech Signals", Wiley-IEEE Press, NY, USA, 1999.
- D. O'Shaughnessy, "Speech Communications: Human and Machine", Second Edition, University Press, 2005.
- T. F. Quatieri, "Discrete time processing of speech signals", Pearson Education, 2005.



Short Term Time Domain Processing

- Need for short term processing of speech
- Compute short term energy and study its significance
- Compute short term zero crossing rate and study its significance
- Compute short term autocorrelation and study its significance.
- Estimate pitch of speech using short term autocorrelation.
- Perform voiced/ unvoiced/ silence classification of speech using short term time domain parameters



Need for Short Term Processing of Speech

- Speech is produced from time-varying system with time-varying excitation
- Therefore speech is a non-stationary signal
- Use of existing signals, systems and processing tools assumes stationarity assumption
- Even if use, the computed values are of little practical importance
- For instance, total energy, $E_T = \sum_{n=-\infty}^{\infty} s^2(n)$ has no significance as speech will have time-varying amplitude and hence energy



Short Term Processing of Speech

- Engineering solution - use existing signal processing tools in a modified fashion.
- Tools assume stationarity, but speech assumed stationary when processed in blocks of 10-30 ms.
- Such a processing is termed as Short Term Processing (STP).
- STP can be performed either in time domain or frequency domain.
- Particular domain depends on information from speech that we are interested in.
- Short term energy, short term zero crossing rate and short term autocorrelation can be computed from time domain.
- Alternatively, short term Fourier transform can be computed from frequency domain.



Short Term Energy (STE)

- How energy is varying with time ?
- Speech signal consist of voiced, unvoiced and silence regions.
- Energy associated with voiced region is large compared to unvoiced region and silence region has least or negligible energy
- STE can be used for voiced, unvoiced and silence classification of speech.

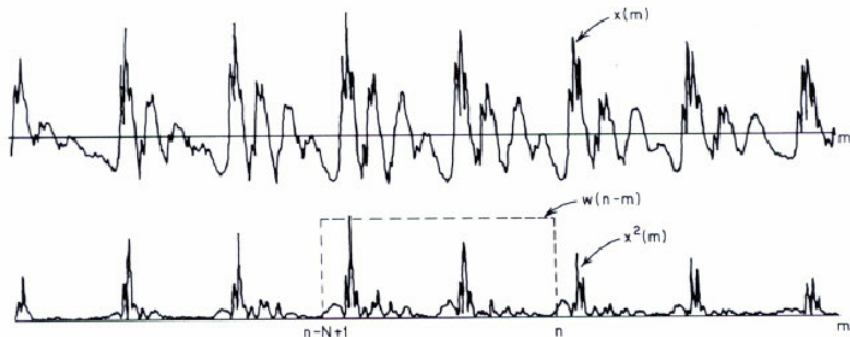


Short Term Energy Equation

- Total energy : $E_T = \sum_{n=-\infty}^{\infty} s^2(n)$
- $E_T = \sum_{n=-\infty}^{-1} s^2(n) + \sum_{n=0}^{(N-1)} s^2(n) + \sum_{n=N}^{\infty} s^2(n)$
- Frame of speech : $n = 0$ to $n = (N - 1)$, where N is frame length, energy of speech zero outside frame length.
- Frame of speech : $s_w(n) = s(m)w(n - m)$, $w(n)$ window fn. of finite duration
- Short term energy frame, $e_f(n) = E_T = \sum_{m=0}^{(N-1)} s^2(m) = \sum_{m=0}^{(N-1)} s_w^2(m) = \sum_{m=0}^{(N-1)} s^2(m)w(n - m)$
- Short term energy $e(n) = \sum_{m=-\infty}^{\infty} s^2(m)w(n - m)$

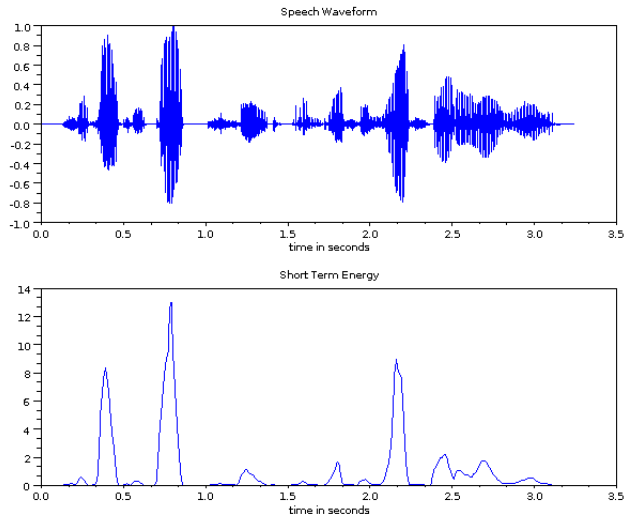


Illustration of STE Computation



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Illustration of STE Contour



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Short Term Zero Crossing Rate (ZCR)

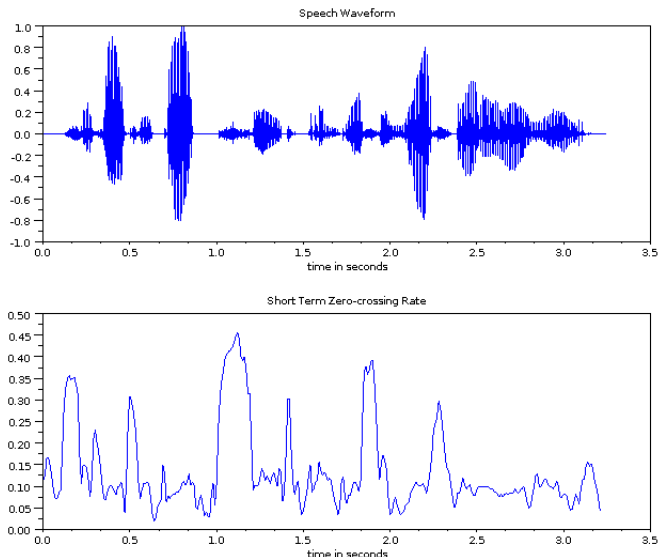
- Speech is a zero mean bipolar signal and hence its amplitude criss-crosses zero amplitude line.
- Zero-crossing rate (ZCR) : number of zero crossings per unit time
- No. of zero crossings more, signal is changing rapidly and may contain high frequency information. Ex. : Unvoiced regions
- No. of zero crossings less, signal changing slowly and signal may contain low frequency information. Ex. : Voiced regions
- ZCR gives an indirect information about frequency content of signal.



Short Term ZCR Equation

- Total energy : $Z_T = \sum_{n=-\infty}^{\infty} |\text{sgn}(s(n)) - \text{sgn}(s(n-1))|$, where signum function, $\text{sgn}(n) = 1$, if $n \geq 0$ and $\text{sgn}(n) = -1$, if $n < 0$
- $Z_T = \sum_{n=-\infty}^{-1} |\text{sgn}(s(n)) - \text{sgn}(s(n-1))| + \sum_{n=0}^{(N-1)} |\text{sgn}(s(n)) - \text{sgn}(s(n-1))| + \sum_{n=N}^{\infty} |\text{sgn}(s(n)) - \text{sgn}(s(n-1))|$
- Frame of speech : $n = 0$ to $n = (N - 1)$, where N is frame length, energy of speech zero outside frame length.
- Frame of speech : $s_w(n) = s(m)w(n - m)$, $w(n)$ window fn.
- Short term ZCR frame,
 $z_f(n) = Z_T = \sum_{m=0}^{(N-1)} |\text{sgn}(s(m)) - \text{sgn}(s(m-1))|w(n - m)$
- Short term ZCR
 $z(n) = \sum_{m=-\infty}^{\infty} |\text{sgn}(s(m)) - \text{sgn}(s(m-1))|w(n - m)$

Illustration of ZCR Contour



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Correlation, Crosscorrelation & Autocorrelation

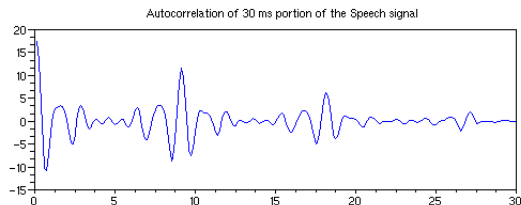
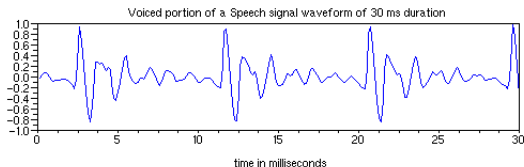
- **Correlation** : What is the relation between given two sequences ?
- Helps in understanding the similarity among the given sequences
- **Crosscorrelation** : Relation or similarity among two different sequences
- **Autocorrelaation** : Relation or similarity of given sequence with itself. Self-similarity
- Same is found out by shifting one sequence at different time lags, multiplying with other sequence and summing the product



Correlation - Equations

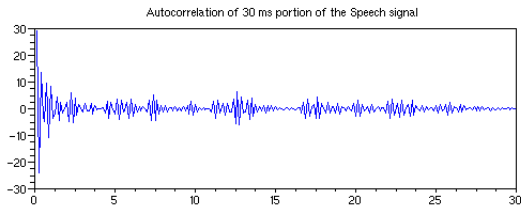
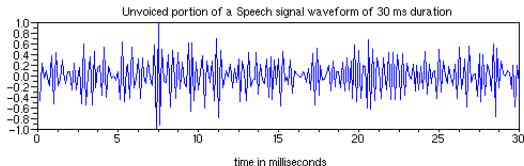
- Let $x(n)$ and $y(n)$ be two sequences, cross-correlation given by
$$r_{xy}(k) = \sum_{n=-\infty}^{\infty} x(n)y(n+k), -\infty \leq k \leq \infty$$
- Let $x(n)$ a sequence, auto-correlation given by
$$r_{xx}(k) = \sum_{n=-\infty}^{\infty} x(n)x(n+k), -\infty \leq k \leq \infty$$
- Let $x_w(n)$ a windowed sequence, short term auto-correlation given by
$$r_{xx}^w(k, n) = \sum_{m=-\infty}^{\infty} x_w(m+n)x_w(m+n+k), -\infty \leq k \leq \infty$$
- Windowed sequence, $x_w(n) = x(n)w(n-n_0)$
- $r_{xx}^w(k, n) = \sum_{m=-\infty}^{\infty} (x(m+n)x(m+n+k))w(n-m),$
$$-\infty \leq k \leq \infty$$

Short Term Autocorrelation - Voiced Segment



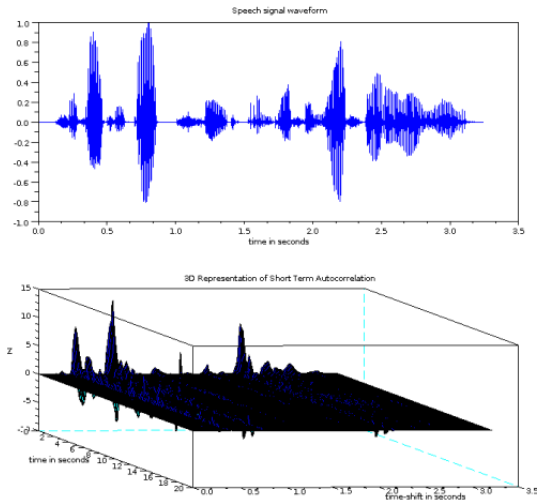
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Short Term Autocorrelation - Unvoiced Segment



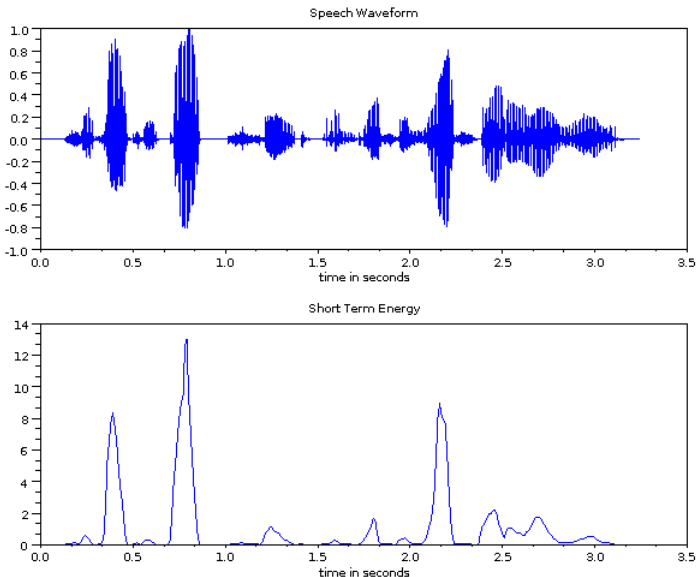
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Short Term Autocorrelation - Speech Signal



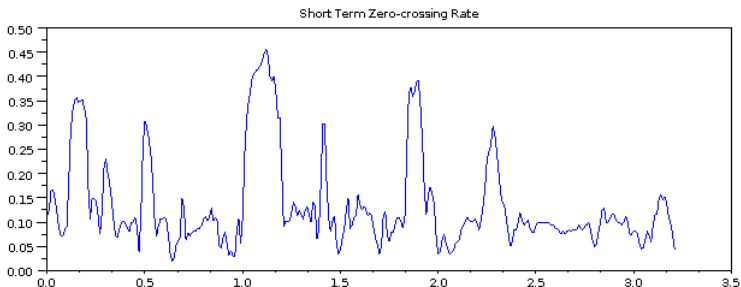
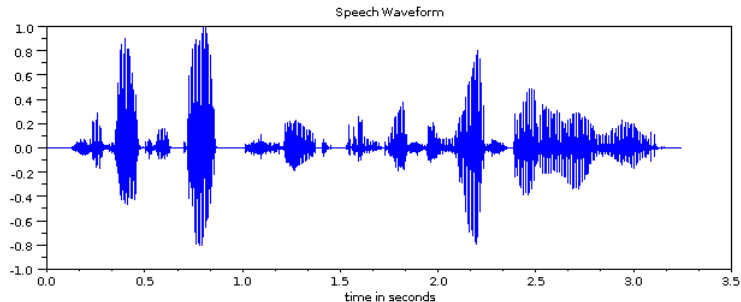
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Short Term Energy Contour

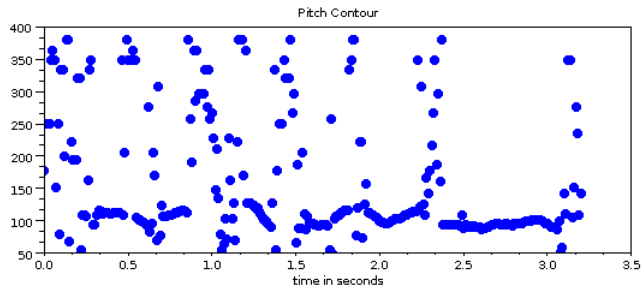
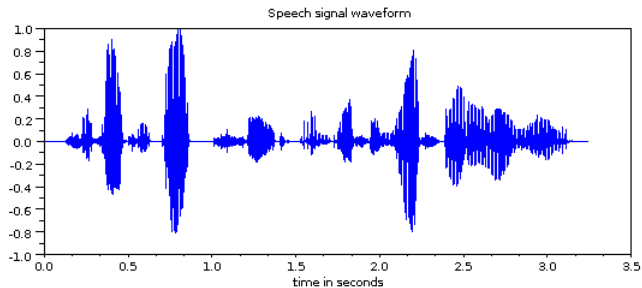


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Short Term ZCR Contour

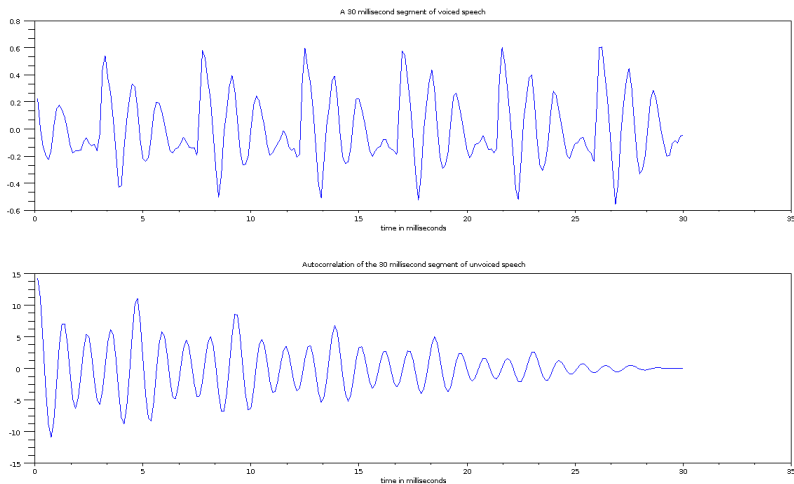


Short Term Pitch Contour



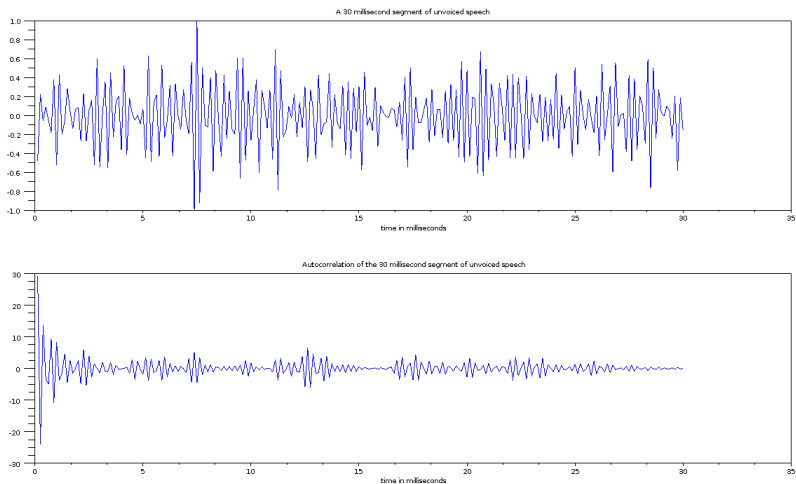
- Voiced / unvoiced /silence
- Voiced : Energy (high / low), ZCR is lowest, pitch is present (periodicity)
- Unvoiced : Energy (low), ZCR is highest, pitch is not present
- Unvoiced : Energy (lowest), ZCR is medium, pitch is not present

Voiced Speech - Autocorrelation



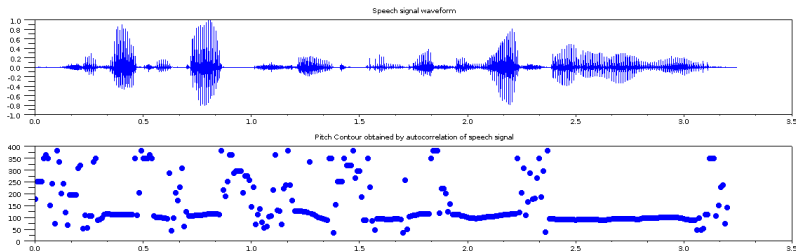
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Unvoiced Speech - Autocorrelation



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Pitch Estimation by Autocorrelation



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